

FLP Seminar 1

$$\text{(IF-TRUE)} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle bl_1, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}$$

$$\text{(IF-FALSE)} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle \quad \langle bl_2, \sigma \rangle \Downarrow \langle \sigma_2 \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \Downarrow \langle \sigma_2 \rangle}$$

$$\text{(WHILE-TRUE)} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle bl, \sigma \rangle \Downarrow \langle \sigma' \rangle \quad \langle \text{while } b \text{ do } bl, \sigma' \rangle \Downarrow \langle \sigma'' \rangle}{\langle \text{while } b \text{ do } bl, \sigma \rangle \Downarrow \langle \sigma'' \rangle}$$

$$\text{(WHILE-FALSE)} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle \text{while } b \text{ do } bl, \sigma \rangle \Downarrow \langle \sigma \rangle}$$

ex 1

$$\sigma(x) = 0$$

$$\sigma(y) = 1$$

$$\sigma(z) = 7$$

+

if $x \leq 7$ then $z := x$ else $z := y$
#

$$(IF-TRUE) \frac{\langle b, \sigma \rangle \Downarrow \langle true \rangle \quad \langle bl_1, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}{\langle \text{if } b \text{ then } bl_1 \text{ else } bl_2, \sigma \rangle \Downarrow \langle \sigma_1 \rangle}$$

Big Step

$$e_{\sigma}(x \leq 7)$$

$$e_{\sigma}(x) \leq e_{\sigma}(7)$$

$$0 \leq 7 = 1$$

$$e_{\sigma}(x \leq 7) = 1 \quad (z := x, \sigma) \Downarrow \sigma_2$$

$$(\text{if } x \leq 7 \text{ then } z := x \text{ else } z := y, \sigma) \Downarrow \sigma_2$$

$$\sigma_2(x) = 0$$

$$\sigma_2(y) = 1$$

$$\sigma_2(z) = 0$$

ex 2

$$\sigma(x) = 6$$

$$\sigma(y) = 3$$

+

while $4 \leq x$ do $x := x - y$

#

$$(\text{WHILE-TRUE}) \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle bl, \sigma \rangle \Downarrow \langle \sigma' \rangle \quad \langle \text{while } b \text{ do } bl, \sigma' \rangle \Downarrow \langle \sigma'' \rangle}{\langle \text{while } b \text{ do } bl, \sigma \rangle \Downarrow \langle \sigma'' \rangle}$$

while $4 \leq x$ do $x := x - y = w$

$$x := x - y, \sigma = bl$$

$$4 \leq x = b$$

$$e_{\sigma}(b)$$

$$e_{\sigma}(4 \leq x)$$

$$e_{\sigma}(4) \leq e_{\sigma}(x)$$

$$4 \leq 6 = 1$$

$$\frac{e_{\sigma}(b) = 1 \quad (bl, \sigma) \Downarrow \sigma_2 \quad (w, \sigma_2) \Downarrow \sigma_2}{(w, \sigma) \Downarrow \sigma_2}$$

$$\sigma_2(x) = 3$$

$$\sigma_2(y) = 3$$

$$e_{\sigma_2}(b) = 0$$

$$\frac{e_{\sigma_2}(b) = 0}{(w, \sigma_2) \Downarrow \sigma_2}$$

ex 3

$$\sigma(x) = 1$$

$$\sigma(y) = 3$$

+

$$\text{while } \neg(y = 0) \text{ do } (y := y - 1; x := 2 * x)$$

#

$$\neg(y = 0) = b$$

$$y := y - 1; x := 2 * x = c$$

$$\text{while } \neg(y = 0) \text{ do } (y := y - 1; x := 2 * x) = w$$

$$e_{\sigma}(b) \quad e_{\sigma}(\neg(y = 0))$$

$$\neg(e_{\sigma}(y) = e_{\sigma}(0))$$

$$\neg(3 = 0) = 1$$

$$e_{\sigma}(b) = 1 \quad (c, \sigma) \Downarrow \sigma_2 \quad (w, \sigma_2) \Downarrow \sigma_3$$

$$(w, \sigma) \Downarrow \sigma_3$$

$$\sigma_2(x) = 2$$

$$\sigma_2(y) = 2$$

$$e_{\sigma_2}(b) = 1$$

$$e_{\sigma_2}(b) = 1 \quad (c, \sigma_2) \Downarrow \sigma_3 \quad (w, \sigma_3) \Downarrow \sigma_4$$

$$(w, \sigma_2) \Downarrow \sigma_4$$

$$\sigma_3(x) = 4$$

$$\sigma_3(y) = 1$$

$$e_{\sigma_3}(b) = 1$$

$$e_{\sigma_3}(b) = 1 \quad (c, \sigma_3) \Downarrow \sigma_4 \quad (w, \sigma_4) \Downarrow \sigma_4$$

$$(w, \sigma_3) \Downarrow \sigma_4$$

$$\sigma_4(x) = 8$$

$$\sigma_4(y) = 0$$

$$e_{\sigma_4}(b) = 0$$

$$e_{\sigma_4}(b) = 0$$

$$(w, \sigma_4) \Downarrow \sigma_4$$

ex 4

$$\sigma(x) = 9$$

$$\sigma(y) = 12$$

+

while $\neg(x = y)$ do (if $x \leq y$ then $y := y - x$
 else $x := x - y$)

#

$$\neg(x = y) = b$$

if $x \leq y$ then $y := y - x$ else $x := x - y = c$

while $\neg(x = y)$ do (if $x \leq y$ then $y := y - x$
 else $x := x - y$) = w

$$e_{\sigma}(\neg(x = y))$$

$$e_{\sigma}(x \leq y) = 1 \quad (y := y - x, \sigma) \Downarrow \sigma_2$$

$$\neg(e_{\sigma}(x) = e_{\sigma}(y))$$

$$(c, \sigma) \Downarrow \sigma_2$$

$$\neg(9 = 12) = 1$$

$$e_{\sigma}(b) = 1 \quad (c, \sigma) \Downarrow \sigma_2 \quad (w, \sigma_2) \Downarrow \sigma_3$$

$$(w, \sigma) \Downarrow \sigma_3$$

$$\sigma_2(x) = 9$$

$$e_{\sigma_2}(x \leq y) = 0 \quad (x := x - y, \sigma_2) \Downarrow \sigma_3$$

$$\sigma_2(y) = 12 - 9 = 3$$

$$(c, \sigma_2) \Downarrow \sigma_3$$

$$e_{\sigma_2}(b) = 1$$

$$e_{\sigma_2}(b) = 1 \quad (c, \sigma) \Downarrow \sigma_3 \quad (w, \sigma_3) \Downarrow \sigma_4$$

$$(w, \sigma_2) \Downarrow \sigma_4$$

$$\sigma_3(x) = 6$$

$$\sigma_3(y) = 3$$

$$e_{\sigma_3}(b) = 1$$

$$e_{\sigma_3}(x \leq y) = 0 \quad (x := x - y, \sigma_3) \Downarrow \sigma_4$$

$$(c, \sigma_3) \Downarrow \sigma_4$$

$$e_{\sigma_3}(b) = 1 \quad (c, \sigma_3) \Downarrow \sigma_4 \quad (w, \sigma_4) \Downarrow \sigma_4$$

$$(w, \sigma_3) \Downarrow \sigma_4$$

$$\sigma_4(x) = 3$$

$$\sigma_4(y) = 3$$

$$e_{\sigma_4}(b) = 0$$

$$e_{\sigma_4}(b) = 0$$

$$(w, \sigma_4) \Downarrow \sigma_4$$